Ibn Tofail University

Analysis V

Problem Set III

Exercise 1:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by

$$f(x,y) = \begin{cases} \dfrac{|xy|^\alpha}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}, \quad \alpha > 0.$$

Determine the values of α for which f is:

- continuous at $(0,0)$.
- differentiable at $(0,0)$.

Correction

Exercise 2:

Study the existence and continuity of the first partial derivatives and the differentiability for the following functions:

$$f(x,y) = \begin{cases} \dfrac{x|y|}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases};$$

$$f(x,y) = \begin{cases} \dfrac{xy}{|x|-y} & \text{if } y \neq |x| \\ 0 & \text{if } y = |x| \end{cases}$$

$$f(x,y) = \begin{cases} \dfrac{x^3-y^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases};$$

$$f(x,y) = \begin{cases} y^2 \sin\left(\dfrac{x}{y}\right) & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases}$$

Correction

Exercise 3:

Calculate $d_{(1,2)}f(x,y)$ and $d_{(2,3,4)}g(x,y,z)$ where $f(x,y) = \dfrac{x}{y^2}$ and $g(x,y,z) = \dfrac{z}{\sqrt{x^2+y^2}}$.

Correction

Exercise 4:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R}^2 \to \mathbb{R}^2$ be the functions defined by

$$f(x,y) = \sin(x^2 - y^2) \quad \text{and} \quad g(x,y) = (x+y, x-y).$$

1. Calculate the gradient of f at $(x,y) \in \mathbb{R}^2$.
2. Calculate the Jacobian matrix of g at $(x,y) \in \mathbb{R}^2$.
3. Deduce the Jacobian matrix of $f \circ g$ at $(x,y) \in \mathbb{R}^2$ and give the expression of $D(f \circ g)(x,y)(h,k)$.

Correction

Exercise 5:

Let $f : \mathbb{R}^n \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} \exp\left(\dfrac{1}{\|x\|^2 - 1}\right) & \text{if } \|x\| < 1 \\ 0 & \text{if } \|x\| \geq 1 \end{cases},$$

where $x = (x_1, \ldots, x_n)$ and $\|x\|^2 = \sum_{i=1}^{n} x_i^2$.

Show that f is of class C^1 on $\mathbb{R}^n$ and give the expression of its differential.

Correction

Exercise 6:

Let f be the function defined by

$$f(x,y) = \begin{cases} xy\dfrac{x^2 - y^2}{x^2 + y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Calculate the partial derivatives at the points $(x,0)$ and $(0,y)$ with $x, y \in \mathbb{R}$. Calculate the second-order partial derivatives at $(0,0)$.

Correction

Exercise 7:

Let f be the function defined by $f(x) = \|x\|$, $\forall x \in \mathbb{R}^n$. Study the differentiability of f on $\mathbb{R}^n$.

Correction